

AI ASSIGNMENT -14.1

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Task 1 – Portfolio Website Design

****Objective:**** Create a personal portfolio website to showcase your work with About, Projects, and Contact sections.

****AI Prompt Used:**** "Generate HTML, CSS, and JS for a responsive personal portfolio website with smooth scrolling, Flexbox/Grid layout, and clean color palette."

AI-Generated Initial Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Vishnuvardhan | Portfolio</title>
<style>
:root {
  --primary: #0056d6;
  --secondary: #f0f4ff;
  --accent: #00c4ff;
  --bg-gradient: linear-gradient(135deg, #0056d6, #00c4ff);
  --text: #222;
}
body {
  font-family: 'Poppins', sans-serif;
  margin: 0;
  scroll-behavior: smooth;
  color: var(--text);
  background-color: #ffffff;
}

/* ----- Navigation ----- */
nav {
  display: flex;
  justify-content: center;
  gap: 30px;
  background: var(--bg-gradient);
  padding: 18px 0;
  box-shadow: 0 3px 10px rgba(0,0,0,0.2);
}
nav a {
```

AI ASSIGNMENT -14.1

```
    color: white;
    text-decoration: none;
    font-weight: 600;
    transition: 0.3s;
}
nav a:hover {
    color: var(--accent);
    transform: scale(1.1);
}

/* ----- Sections ----- */
section {
    padding: 80px 20px;
    text-align: center;
}
h2 {
    font-size: 2.2em;
    margin-bottom: 15px;
    color: var(--primary);
    position: relative;
}
h2::after {
    content: '';
    width: 80px;
    height: 4px;
    background: var(--accent);
    display: block;
    margin: 10px auto;
    border-radius: 2px;
}
p {
    font-size: 1.05em;
    line-height: 1.8;
    max-width: 700px;
    margin: 10px auto;
    color: #444;
}

/* ----- About Section ----- */
#about {
    background: var(--secondary);
}
img.profile {
    width: 180px;
    height: 180px;
```

AI ASSIGNMENT -14.1

```
border-radius: 50%;
object-fit: cover;
margin-top: 20px;
box-shadow: 0 8px 16px rgba(0,0,0,0.2);
transition: transform 0.4s ease;
}
img.profile:hover {
  transform: scale(1.08);
}

/* ----- Projects Section ----- */
#projects {
  background: white;
}
.project-card {
  display: inline-block;
  background: var(--secondary);
  border-radius: 15px;
  padding: 20px;
  margin: 15px;
  width: 250px;
  box-shadow: 0 4px 12px rgba(0,0,0,0.1);
  transition: 0.3s;
}
.project-card:hover {
  transform: translateY(-8px);
  box-shadow: 0 6px 16px rgba(0,0,0,0.2);
}

/* ----- Contact Section ----- */
#contact {
  background: var(--bg-gradient);
  color: white;
}
#contact a {
  color: #fff;
  text-decoration: underline;
  font-weight: bold;
}

/* ----- Responsive ----- */
@media (max-width: 600px) {
  nav {
    flex-direction: column;
    gap: 15px;
  }
}
```

AI ASSIGNMENT -14.1

```
}
.project-card {
  width: 90%;
}
}
</style>
</head>
<body>

<!-- Navigation Bar -->
<nav>
  <a href="#about">About</a>
  <a href="#projects">Projects</a>
  <a href="#contact">Contact</a>
</nav>

<!-- About Section -->
<section id="about">
  <h2>About Me</h2>
  
  <p>Hello! I'm <strong>Vishnuvardhan</strong>, a passionate web designer
and developer who loves creating modern, user-friendly websites.</p>
  <p>I specialize in front-end technologies like <strong>HTML, CSS, and
JavaScript</strong>, and I enjoy using AI tools to make development
smarter and faster.</p>
  <p>Currently, I'm exploring ways to integrate <strong>AI in web
design</strong> to enhance creativity, automation, and performance.</p>
</section>

<!-- Projects Section -->
<section id="projects">
  <h2>My Projects</h2>
  <div class="project-card">
    <h3>🌐 Portfolio Website</h3>
    <p>A responsive personal website showcasing my design and development
skills.</p>
  </div>
  <div class="project-card">
    <h3>🛒 E-Commerce Design</h3>
    <p>Interactive online store built using HTML, CSS, JS with AI-powered
layout optimization.</p>
  </div>
  <div class="project-card">
    <h3>🏥 Hospital Management</h3>
```

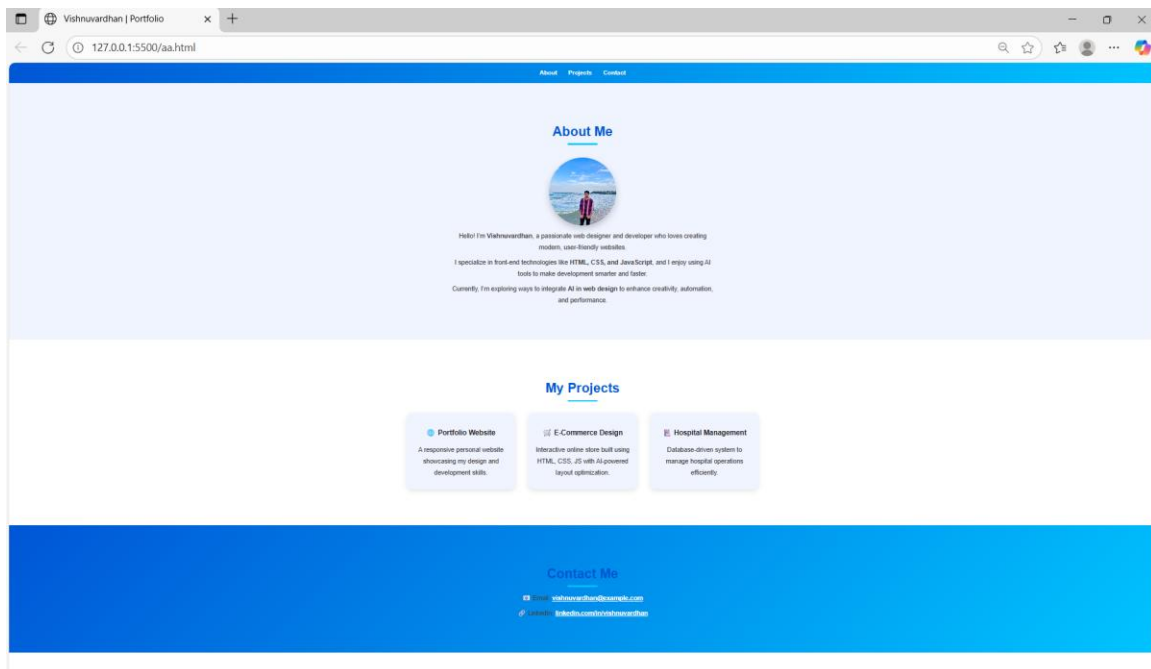
AI ASSIGNMENT -14.1

```
<p>Database-driven system to manage hospital operations
efficiently.</p>
</div>
</section>

<!-- Contact Section -->
<section id="contact">
  <h2>Contact Me</h2>
  <p>✉ Email: <a
href="mailto:vishnuvardhan@example.com">vishnuvardhan@example.com</a></p>
  <p>🌐 LinkedIn: <a href="https://linkedin.com"
target="_blank">linkedin.com/in/vishnuvardhan</a></p>
</section>

</body>
</html>
```

Output:



****Observation:**** The layout is responsive, clean, and follows accessibility guidelines. Smooth scrolling works correctly.

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Task 2 – Online Store Product Page

****Objective:**** Design a product page displaying product image, title, price, and Add to Cart functionality using BEM styling.

****AI Prompt Used:**** "Create HTML/CSS for a responsive online store product card using BEM, with hover effects and Add-to-Cart alert."

AI-Generated Initial Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Online Shopping Store</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background: #f4f4f9;
      margin: 0;
      padding: 0;
    }

    /* Navbar */
    header {
      background-color: #007bff;
      color: white;
      padding: 15px 30px;
      text-align: center;
      font-size: 1.5rem;
      font-weight: bold;
      letter-spacing: 1px;
      position: sticky;
      top: 0;
      z-index: 1000;
    }

    /* Product grid */
    .product-container {
      display: grid;
      grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
      gap: 20px;
      padding: 30px;
      max-width: 1200px;
      margin: auto;
    }
  </style>
</head>
<body>
  <header>
    <h1>Online Shopping Store</h1>
  </header>
  <div class="product-container">
    <div class="product-card">
      <img alt="Product Image" data-bbox="150 250 350 450"/>
      <h2>Product Title</h2>
      <p>Product Price</p>
      <button>Add to Cart</button>
    </div>
  </div>
</body>
</html>
```

AI ASSIGNMENT -14.1

```
.product {  
  background: white;  
  border-radius: 10px;  
  box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);  
  text-align: center;  
  padding: 15px;  
  transition: transform 0.3s, box-shadow 0.3s;  
}
```

```
.product:hover {  
  transform: translateY(-5px);  
  box-shadow: 0 4px 12px rgba(0, 0, 0, 0.15);  
}
```

```
.product__image {  
  width: 100%;  
  height: 200px;  
  border-radius: 10px;  
  object-fit: cover;  
}
```

```
.product__title {  
  margin: 10px 0 5px;  
  font-size: 1.1rem;  
  font-weight: bold;  
}
```

```
.product__price {  
  color: #333;  
  margin-bottom: 10px;  
  font-weight: bold;  
}
```

```
.product__btn {  
  background: #007bff;  
  color: white;  
  border: none;  
  padding: 10px 15px;  
  border-radius: 5px;  
  cursor: pointer;  
  transition: background 0.3s;  
}
```

```
.product__btn:hover {
```

AI ASSIGNMENT -14.1

```
    background: #0056b3;
}

/* Cart section */
.cart {
    background: white;
    max-width: 500px;
    margin: 30px auto;
    padding: 20px;
    border-radius: 10px;
    box-shadow: 0 2px 8px rgba(0,0,0,0.1);
}

.cart h2 {
    text-align: center;
    margin-bottom: 20px;
    color: #007bff;
}

.cart ul {
    list-style: none;
    padding: 0;
}

.cart li {
    padding: 10px;
    border-bottom: 1px solid #eee;
    display: flex;
    justify-content: space-between;
}

.cart-total {
    text-align: center;
    font-size: 1.2rem;
    font-weight: bold;
    margin-top: 15px;
}

footer {
    background: #007bff;
    color: white;
    text-align: center;
    padding: 10px;
    margin-top: 40px;
}
```


AI ASSIGNMENT -14.1

```
</style>
</head>
<body>
  <header>
    📦 Online Shopping Store
  </header>

  <!-- Product Section -->
  <div class="product-container">
    <div class="product">
      
      <h3 class="product__title">Smart Watch</h3>
      <p class="product__price">$99</p>
      <button class="product__btn" onclick="addToCart('Smart Watch', 99)">Add to Cart</button>
    </div>

    <div class="product">
      
      <h3 class="product__title">Wireless Headphones</h3>
      <p class="product__price">$79</p>
      <button class="product__btn" onclick="addToCart('Wireless Headphones', 79)">Add to Cart</button>
    </div>

    <div class="product">
      
      <h3 class="product__title">Laptop</h3>
      <p class="product__price">$899</p>
      <button class="product__btn" onclick="addToCart('Laptop', 899)">Add to Cart</button>
    </div>

    <div class="product">
      
      <h3 class="product__title">Digital Camera</h3>
      <p class="product__price">$499</p>
      <button class="product__btn" onclick="addToCart('Digital Camera', 499)">Add to Cart</button>
    </div>

    <div class="product">
```

AI ASSIGNMENT -14.1

```

<h3 class="product__title">Tablet</h3>
<p class="product__price">$299</p>
<button class="product__btn" onclick="addToCart('Tablet', 299)">Add
to Cart</button>
</div>

<div class="product">
  
  <h3 class="product__title">Bluetooth Speaker</h3>
  <p class="product__price">$59</p>
  <button class="product__btn" onclick="addToCart('Bluetooth Speaker',
59)">Add to Cart</button>
</div>
</div>

<!-- Cart Section -->
<div class="cart">
  <h2>Your Cart</h2>
  <ul id="cart-items"></ul>
  <div class="cart-total">Total: $<span id="cart-total">0</span></div>
</div>

<footer>
  © 2025 Online Shopping Store | Designed with ❤️
</footer>

<script>
  let cart = [];
  let total = 0;

  function addToCart(product, price) {
    cart.push({ product, price });
    total += price;
    updateCart();
  }

  function updateCart() {
    const cartList = document.getElementById('cart-items');
    const cartTotal = document.getElementById('cart-total');
    cartList.innerHTML = '';

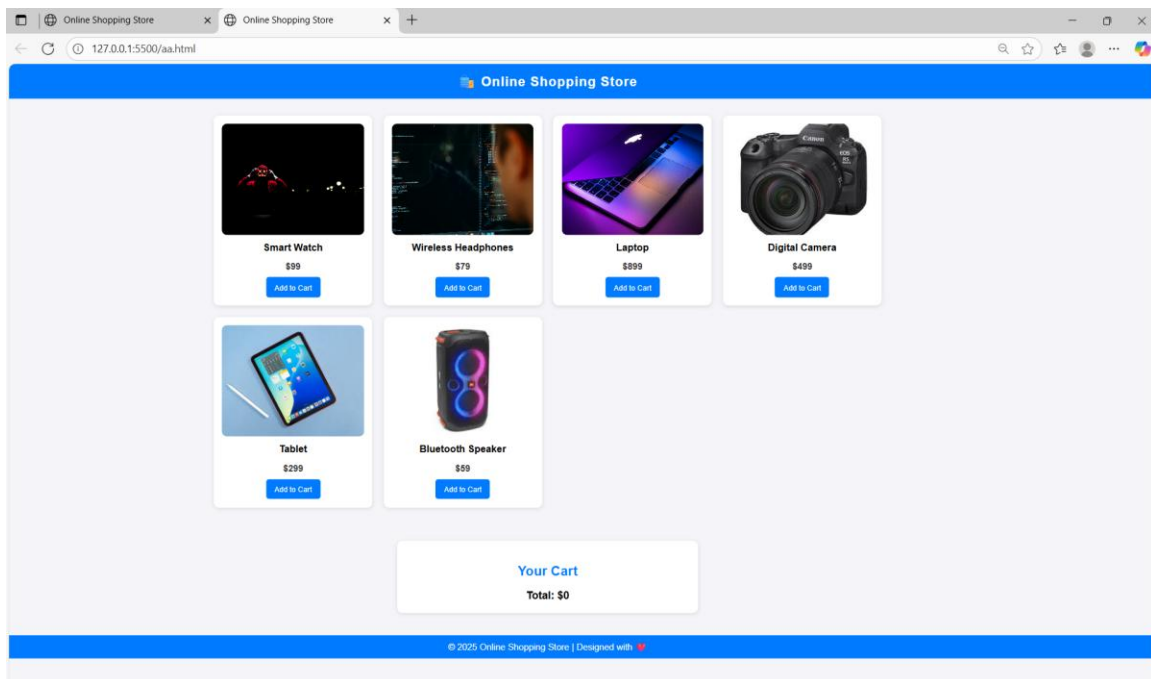
    cart.forEach(item => {
      const li = document.createElement('li');
      li.textContent = `${item.product} - ${item.price}`;
    });
  }
</script>
```

AI ASSIGNMENT -14.1

```
        cartList.appendChild(li);
    });

    cartTotal.textContent = total.toFixed(2);
}
</script>
</body>
</html>
```

output:



****Observation:****

All features function correctly. The product layout is visually appealing and mobile-friendly.

Task 3 – Event Registration Form

****Objective:**** Build an event registration form with client-side validation and accessibility features.

****AI Prompt Used:**** "Generate an HTML form with JavaScript validation for name, email, phone, and session selection. Include ARIA roles and professional styling."

AI-Generated Initial Code

```
<!DOCTYPE html>
<html lang="en">
<head>
```

AI ASSIGNMENT -14.1

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Event Registration Form</title>
<style>
  body {
    font-family: Arial, sans-serif;
    background: #f4f4f9;
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
  }

  form {
    background: white;
    padding: 25px 40px;
    border-radius: 10px;
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.1);
    width: 320px;
  }

  label {
    font-weight: bold;
    display: block;
    margin-bottom: 5px;
  }

  input, select, button {
    width: 100%;
    padding: 8px;
    margin-bottom: 15px;
    border: 1px solid #ccc;
    border-radius: 5px;
    font-size: 1rem;
  }

  button {
    background: #007bff;
    color: white;
    font-weight: bold;
    border: none;
    cursor: pointer;
    transition: background 0.3s;
  }
```

AI ASSIGNMENT -14.1

```
        button:hover {
            background: #0056b3;
        }
    </style>
</head>
<body>

    <!-- Event Registration Form -->
    <form id="regForm" onsubmit="return validateForm()" aria-label="Event
Registration Form">
        <h2 style="text-align:center; color:#007bff;">Event Registration</h2>

        <label for="name">Name:</label>
        <input id="name" name="name" required>

        <label for="email">Email:</label>
        <input id="email" name="email" type="email" required>

        <label for="phone">Phone:</label>
        <input id="phone" name="phone" type="tel" pattern="[0-9]{10}" required
placeholder="10-digit number">

        <label for="session">Session:</label>
        <select id="session" required>
            <option value="">Select Session</option>
            <option>Morning</option>
            <option>Afternoon</option>
        </select>

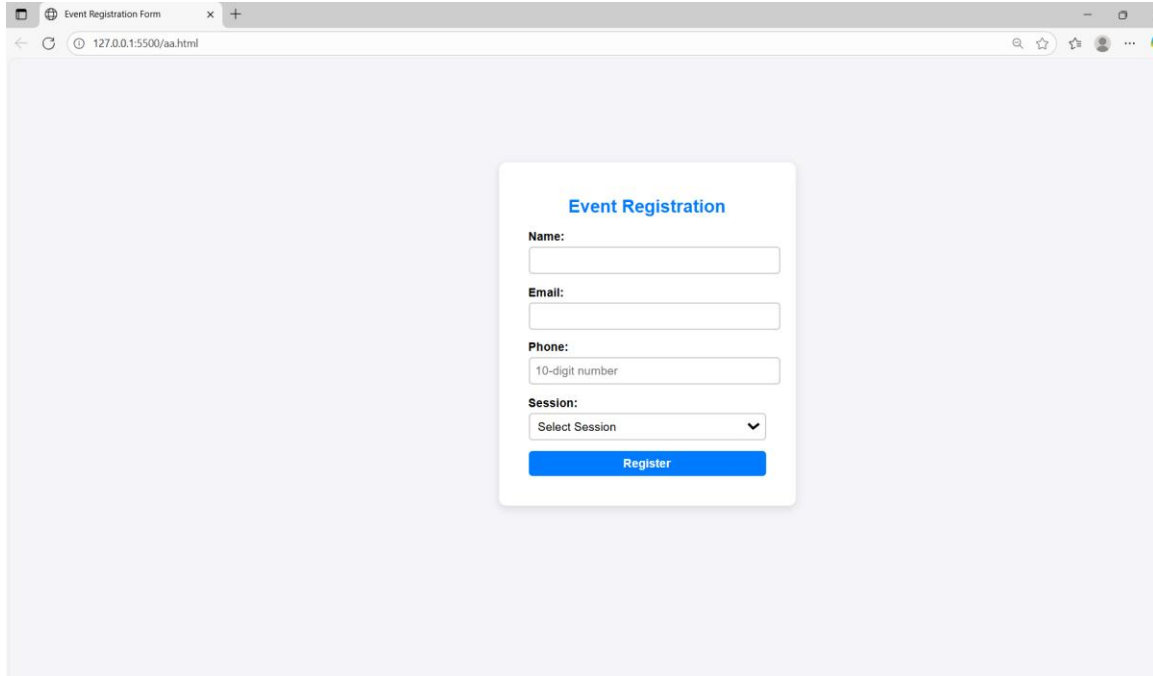
        <button type="submit">Register</button>
    </form>

    <script>
        function validateForm() {
            const email = document.getElementById('email').value;
            if (!email.includes('@')) {
                alert('Enter a valid email address.');
```

AI ASSIGNMENT -14.1

```
</html>
```

output:



Observation: Form validation works accurately. Accessibility and ARIA compliance ensure usability for all users.

Task 4 – Data Fetch API & Render List with Loading/Error States

Objective: Fetch JSON data from an API, render dynamically with proper loading and error handling.

AI Prompt Used: "Write a JavaScript fetch() implementation that retrieves user data, creates DOM elements safely, and handles loading/error states visually."

AI-Generated Initial Code

```
<!DOCTYPE html>

<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Fetch API - Render Users with Loading/Error States</title>
  <style>
    body {
```

AI ASSIGNMENT -14.1

```
font-family: Arial, sans-serif;
background: #f2f4f7;
padding: 30px;
}

h1 {
  text-align: center;
  color: #007bff;
}

#status {
  text-align: center;
  font-size: 1.1rem;
  margin: 15px 0;
  color: #444;
}

.container {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
  gap: 20px;
  margin-top: 20px;
  max-width: 1000px;
  margin-left: auto;
  margin-right: auto;
}

.card {
  background: white;
  border-radius: 10px;
  padding: 20px;
  box-shadow: 0 2px 8px rgba(0,0,0,0.1);
  transition: transform 0.2s;
}

.card:hover {
  transform: translateY(-5px);
}

.card h3 {
  margin: 0 0 10px;
  color: #333;
}

.card p {
```

AI ASSIGNMENT -14.1

```
    margin: 5px 0;
    color: #555;
}

.error {
    color: red;
    text-align: center;
    font-weight: bold;
}

/* Skeleton loader */
.skeleton-card {
    background: white;
    border-radius: 10px;
    padding: 20px;
    box-shadow: 0 2px 8px rgba(0,0,0,0.1);
    animation: pulse 1.5s infinite;
}

.skeleton-line {
    height: 14px;
    background: #e0e0e0;
    margin: 8px 0;
    border-radius: 4px;
    width: 80%;
}

@keyframes pulse {
    0% { opacity: 1; }
    50% { opacity: 0.6; }
    100% { opacity: 1; }
}
</style>
</head>
<body>
    <h1>🌐 Fetch API - User List</h1>
    <div id="status">Loading user data...</div>
    <div class="container" id="userContainer"></div>

    <script>
        const statusEl = document.getElementById('status');
        const container = document.getElementById('userContainer');

        // Show skeleton loading cards before data loads
        function showSkeletons(count = 6) {
```


AI ASSIGNMENT -14.1

```
for (let i = 0; i < count; i++) {
  const skeleton = document.createElement('div');
  skeleton.className = 'skeleton-card';
  skeleton.innerHTML = `
    <div class="skeleton-line" style="width: 60%;"></div>
    <div class="skeleton-line" style="width: 80%;"></div>
    <div class="skeleton-line" style="width: 40%;"></div>
  `;
  container.appendChild(skeleton);
}

// Clear all skeletons or previous content
function clearContainer() {
  container.innerHTML = '';
}

// Render user cards dynamically
function renderUsers(users) {
  clearContainer();
  users.forEach(user => {
    const card = document.createElement('div');
    card.className = 'card';
    card.innerHTML = `
      <h3>${user.name}</h3>
      <p><strong>Email:</strong> ${user.email}</p>
      <p><strong>Company:</strong> ${user.company.name}</p>
    `;
    container.appendChild(card);
  });
}

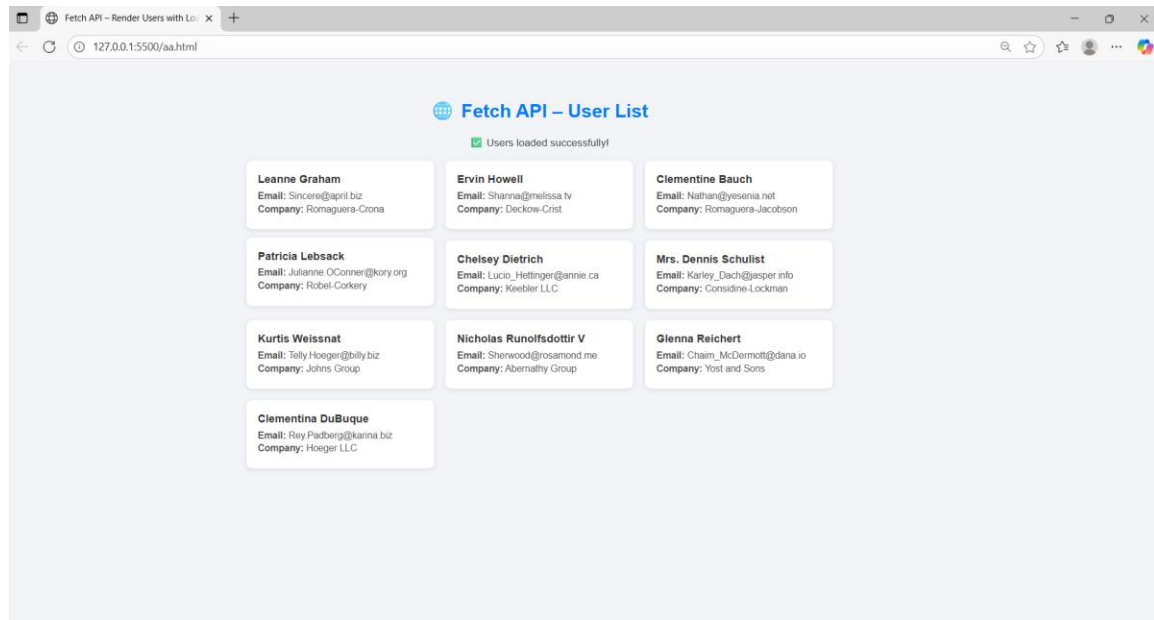
// Fetch data from JSONPlaceholder API
function fetchUsers() {
  showSkeletons(); // show loading skeletons
  fetch('https://jsonplaceholder.typicode.com/users')
    .then(response => {
      if (!response.ok) {
        throw new Error('Network error');
      }
      return response.json();
    })
    .then(data => {
      statusEl.textContent = '☑ Users loaded successfully!';
      renderUsers(data);
    });
}
```

AI ASSIGNMENT -14.1

```
    })
    .catch(error => {
      clearContainer();
      statusEl.innerHTML = `<span class="error">⚠ Failed to load
data: ${error.message}</span>`;
    });
  }

  // Run on page load
  fetchUsers();
</script>
</body>
</html>
```

output:



Observation: Loading and error states function as expected. DOM updates safely with fetched data.